The effect of inhaling peppermint odor and ethanol in women athletes.

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The purpose of this study was to determine whether inhaling peppermint odor has effects on time of running, maximum heart rate (MHR), maximum oxygen consumption (VO2max), oxygen consumption (VO2), minute ventilation (VE) and respiratory exchange ratio (RER) during acute intensive exercise or not. 36 women soccer player were chosen for participating in this research. They were randomly divided in 3 groups (control, inhaling peppermint, inhaling mixture of peppermint and ethanol). In order to be aware of similarity of groups, the subjects' BMI was determined and ANOVA did not show any significant differences ($p < 0.05$). The subjects of three groups ran on treadmill according to Bruce test. Heart rate, time of running, VO2max, VO2, VE and RER were measured by Gas Analyzer. After collecting the data, ANOVA was done ($p < 0.05$) and the results showed that in this study the inhaling of fragrant odors did not have any significant effect on the time of running, MHR, VO2max, VO2, VE and RER, which we think is due to the intensity and duration of training. Referring to our results of the present study; we suggest that inhaling peppermint odor during acute intensive exercise has no significant effect on pulmonary indexes and physical performance (Tab. 4, Fig. 1, Ref. 21).